

Statistics and Probability: Random Variables and Sampling

Every day, we encounter situations involving uncertainty. Will it rain tomorrow? How many customers will enter a store today? What is the average height of Grade 11 students in a school? These questions involve **random variables** – quantities whose values depend on chance. Understanding random variables allows us to describe, predict, and make decisions about uncertain events. In statistics, we also rely on **sampling** – studying a small part of a population to draw conclusions about the whole. Sampling makes research practical when studying every individual is impossible or too costly.

This reading material differentiates discrete and continuous random variables and describes their uses. It explains how to calculate the expected value, variance, and standard deviation of discrete random variables. It illustrates the normal random variable and its characteristics, shows how to convert a normal random variable to a standard normal variable and vice versa, and demonstrates how to compute probabilities and percentiles using the Z-distribution table. Finally, it explains random sampling, how to determine appropriate sample size, and distinguishes between parameter and statistic, and between probability and nonprobability sampling techniques.

Part 1: Discrete and Continuous Random Variables

What Is a Random Variable?

A **random variable** is a variable whose value depends on the outcome of a random event. It assigns a numerical value to each outcome of an experiment.

Example: When you roll a die, the outcome is random. The random variable X could represent the number that appears (1, 2, 3, 4, 5, or 6).

Discrete vs. Continuous Random Variables

Feature	Discrete Random Variable	Continuous Random Variable
Definition	A variable that can take on only specific, countable values	A variable that can take on any value within a range or interval
Values	Whole numbers, counted values	Measured values, including decimals and fractions
Examples	Number of students in a class; number of cars in a parking lot; number of heads in 3 coin flips	Height of students; weight of newborns; temperature; time to finish a race
Representation	Usually counted (0, 1, 2, 3, ...)	Usually measured (any value within a range)
Graph	Bar graph or histogram with gaps between bars	Smooth curve (no gaps)

Uses of Discrete Random Variables

Use	Example
Counting events	Number of defective items in a batch
Probability of success	Number of correct answers on a test
Quality control	Number of product defects per day
Games and lotteries	Number of winning tickets

Uses of Continuous Random Variables

Use	Example
Measuring physical quantities	Height, weight, temperature
Time measurements	Waiting time, travel time
Financial data	Stock prices, income
Scientific research	Concentration of chemicals, speed of particles

Probability Distribution of a Discrete Random Variable

A **probability distribution** lists all possible values of a discrete random variable and their corresponding probabilities.

Example: Tossing Two Coins

Let X = number of heads.

X (Number of Heads)	Probability $P(X)$
0	$1/4 = 0.25$
1	$2/4 = 0.50$
2	$1/4 = 0.25$
Total	1.00

Key rule: The sum of all probabilities must equal 1.

Part 2: Expected Value, Variance, and Standard Deviation

Expected Value (Mean) of a Discrete Random Variable

The **expected value** (also called the mean) is the average value of the random variable over many trials. It is denoted by **E(X)** or **μ**.

$$E(X) = \sum [x \cdot P(x)]$$

Where:

- x = each possible value of the random variable
- $P(x)$ = probability of that value

Example 1: Expected Value

Using the two-coin example:

X	P(X)	X · P(X)
0	0.25	0
1	0.50	0.50
2	0.25	0.50
Total	1.00	1.00

$$E(X) = 0 + 0.50 + 0.50 = \mathbf{1.00}$$

Answer: The expected number of heads is **1.0**.

Variance of a Discrete Random Variable

The **variance** measures how spread out the values are from the mean. It is denoted by **Var(X)** or σ^2 .

$$\text{Var}(X) = \Sigma [(x - \mu)^2 \cdot P(x)]$$

An easier computational formula:

$$\text{Var}(X) = \Sigma [x^2 \cdot P(x)] - \mu^2$$

Example 2: Variance

Using the same example with $\mu = 1.0$:

X	P(X)	X^2	$X^2 \cdot P(X)$
0	0.25	0	0
1	0.50	1	0.50
2	0.25	4	1.00
Total	1.00		1.50

$$\text{Var}(X) = 1.50 - (1.0)^2 = 1.50 - 1.00 = \mathbf{0.50}$$

Standard Deviation of a Discrete Random Variable

The **standard deviation** is the square root of the variance. It is denoted by σ .

$$\sigma = \sqrt{\text{Var}(X)}$$

Example 3: Standard Deviation

$$\sigma = \sqrt{0.50} \approx \mathbf{0.707}$$

Answer: The standard deviation is approximately **0.71**.

Summary of Formulas

Measure	Formula
Expected Value (Mean)	$E(X) = \sum [x \cdot P(x)]$
Variance	$\text{Var}(X) = \sum [x^2 \cdot P(x)] - \mu^2$
Standard Deviation	$\sigma = \sqrt{\text{Var}(X)}$

Part 3: The Normal Random Variable

What Is a Normal Random Variable?

A **normal random variable** is a continuous random variable that follows a **normal distribution** – a bell-shaped curve that is symmetric about the mean. Many natural phenomena follow a normal distribution: heights, weights, test scores, and measurement errors.

Characteristics of the Normal Distribution

Characteristic	Description
Bell-shaped	The curve is highest at the center and tapers off at both ends
Symmetric	The left half is a mirror image of the right half
Mean = Median = Mode	All three measures of central tendency are equal and located at the center
Total area = 1	The area under the curve represents total probability
Asymptotic	The curve approaches but never touches the horizontal axis
Defined by μ and σ	The mean (μ) determines the center; the standard deviation (σ) determines the spread

The Empirical Rule (68-95-99.7 Rule)

Range	Percentage of Data
$\mu \pm 1\sigma$	Approximately 68%
$\mu \pm 2\sigma$	Approximately 95%
$\mu \pm 3\sigma$	Approximately 99.7%

Illustration of the Normal Curve

The normal curve is centered at the mean μ . The standard deviation σ controls how wide or narrow the curve is.

- Small $\sigma \rightarrow$ tall, narrow curve (data clustered near mean)
 - Large $\sigma \rightarrow$ short, wide curve (data spread out)
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Part 4: Standard Normal Variable and Z-Scores

What Is the Standard Normal Distribution?

The **standard normal distribution** is a special normal distribution with:

$$\mu = 0 \text{ and } \sigma = 1$$

Any normal random variable can be converted to a standard normal variable using the **z-score formula**.

The Z-Score Formula

$$z = (x - \mu) / \sigma$$

Where:

- x = the value being standardized
- μ = mean of the distribution
- σ = standard deviation of the distribution

Converting Normal to Standard Normal

Example 1: Finding the Z-Score

A test has a mean of 75 and a standard deviation of 8. A student scores 85. Find the z-score.

$$z = (85 - 75) / 8 = 10 / 8 = \mathbf{1.25}$$

Answer: The z-score is **1.25**, meaning the score is 1.25 standard deviations above the mean.

Converting Standard Normal to Normal

Example 2: Finding the Raw Score

Using the same distribution ($\mu = 75$, $\sigma = 8$), find the raw score for $z = -0.5$.

$$\begin{aligned}x &= \mu + z\sigma \\x &= 75 + (-0.5)(8) \\x &= 75 - 4 \\x &= \mathbf{71}\end{aligned}$$

Answer: The raw score is **71**.

Summary of Conversions

Conversion	Formula
Normal to Standard Normal	$z = (x - \mu) / \sigma$
Standard Normal to Normal	$x = \mu + z\sigma$

Part 5: Computing Probabilities and Percentiles Using the Z-Table

How to Use the Z-Table

The **Z-table** (also called the standard normal table) gives the area under the curve to the **left** of a given z-score. This area represents probability.

Common Probability Notations

Notation	Meaning
$P(Z < z)$	Probability that z is less than a given value (area to the left)
$P(Z > z)$	Probability that z is greater than a given value (area to the right)
$P(a < Z < b)$	Probability between two values

Key Z-Table Values to Memorize

z	$P(Z < z)$
-3.00	0.0013
-2.00	0.0228
-1.96	0.0250

z	$P(Z < z)$
-1.00	0.1587
0.00	0.5000
1.00	0.8413
1.96	0.9750
2.00	0.9772
3.00	0.9987

Example 1: Probability Less Than a Value

The heights of Grade 11 students are normally distributed with $\mu = 165$ cm and $\sigma = 6$ cm. Find the probability that a randomly selected student is shorter than 170 cm.

Step 1: Find the z-score.

$$z = (170 - 165) / 6 = 5 / 6 \approx 0.83$$

Step 2: Use the Z-table.

$$P(Z < 0.83) \approx \mathbf{0.7967}$$

Answer: The probability is approximately **0.7967** or **79.67%**.

Example 2: Probability Greater Than a Value

Using the same distribution, find the probability that a student is taller than 175 cm.

Step 1: Find the z-score.

$$z = (175 - 165) / 6 = 10 / 6 \approx 1.67$$

Step 2: Find $P(Z > 1.67)$.

$$P(Z < 1.67) = 0.9525$$

$$P(Z > 1.67) = 1 - 0.9525 = \mathbf{0.0475}$$

Answer: The probability is approximately **0.0475** or **4.75%**.

Example 3: Probability Between Two Values

Find the probability that a student's height is between 160 cm and 172 cm.

Step 1: Find both z-scores.

$$z_1 = (160 - 165) / 6 = -0.83$$

$$z_2 = (172 - 165) / 6 \approx 1.17$$

Step 2: Find the area between.

$$P(Z < -0.83) \approx 0.2033$$

$$P(Z < 1.17) \approx 0.8790$$

$$P(-0.83 < Z < 1.17) = 0.8790 - 0.2033 = \mathbf{0.6757}$$

Answer: The probability is approximately **0.6757** or **67.57%**.

Finding Percentiles

A **percentile** is the value below which a given percentage of data falls.

Example 4: Finding the 90th Percentile

Find the height that corresponds to the 90th percentile.

Step 1: Find the z-score for 0.90.

$$P(Z < z) = 0.90 \rightarrow z \approx 1.28$$

Step 2: Convert to raw score.

$$x = \mu + z\sigma = 165 + (1.28)(6) = 165 + 7.68 = \mathbf{172.68 \text{ cm}}$$

Answer: The 90th percentile height is approximately **172.68 cm**.

Part 6: Solving Problems Involving Normal Distribution

Example 1: Test Scores

The scores on a standardized test are normally distributed with $\mu = 500$ and $\sigma = 100$. What percentage of students scored above 650?

$$z = (650 - 500) / 100 = 1.50$$

$$P(Z > 1.50) = 1 - P(Z < 1.50) = 1 - 0.9332 = \mathbf{0.0668}$$

Answer: Approximately **6.68%** of students scored above 650.

Example 2: Quality Control

The weights of bags of rice are normally distributed with $\mu = 50$ kg and $\sigma = 2$ kg. A bag is considered underweight if it weighs less than 47 kg. What percentage of bags are underweight?

$$z = (47 - 50) / 2 = -1.50$$

$$P(Z < -1.50) = \mathbf{0.0668}$$

Answer: Approximately **6.68%** of bags are underweight.

Example 3: Finding a Cutoff Value

Using the rice example, find the weight below which only 5% of bags fall.

Step 1: Find the z-score for 0.05.

$$P(Z < z) = 0.05 \rightarrow z \approx -1.645$$

Step 2: Convert to raw score.

$$x = 50 + (-1.645)(2) = 50 - 3.29 = \mathbf{46.71 \text{ kg}}$$

Answer: Only 5% of bags weigh less than **46.71 kg**.

Part 7: Random Sampling

What Is Random Sampling?

Random sampling is a method of selecting a sample from a population where every member of the population has an equal chance of being selected. It ensures that the sample is representative of the population.

Why Random Sampling Matters

Reason	Explanation
Reduces bias	Every member has an equal chance; no group is favored
Increases validity	Results can be generalized to the population
Allows statistical inference	Probability theory can be applied to estimate population parameters
Supports fairness	Used in lotteries, surveys, and clinical trials

Common Random Sampling Techniques

Technique	Description	Example
Simple random sampling	Every member has an equal chance; selection is completely random	Drawing names from a hat
Systematic sampling	Select every kth member from a list	Every 10th student on a class list
Stratified sampling	Divide population into groups (strata), then randomly sample from each group	Selecting 5 boys and 5 girls from each class
Cluster sampling	Divide population into clusters, then randomly select entire clusters	Randomly selecting 5 barangays, then surveying all residents

Part 8: Determining Sample Size

Why Sample Size Matters

Too Small

Results may be unreliable

High margin of error

Cannot detect real differences

Too Large

Wastes time, money, and resources

May be unnecessary if smaller sample suffices

May inconvenience participants

Slovin's Formula

$$n = N / (1 + Ne^2)$$

Where:

- n = sample size
- N = population size
- e = margin of error (usually 0.05 or 5%)

Example 1: Using Slovin's Formula

A school has 2,000 students. Find the sample size needed for a 5% margin of error.

$$n = 2000 / (1 + 2000 \times 0.05^2)$$

$$n = 2000 / (1 + 2000 \times 0.0025)$$

$$n = 2000 / (1 + 5)$$

$$n = 2000 / 6$$

$$n \approx 333.33$$

Answer: The sample size should be **334 students**.

Example 2: Different Margin of Error

Using the same population ($N = 2,000$) with a 3% margin of error:

$$n = 2000 / (1 + 2000 \times 0.03^2)$$

$$n = 2000 / (1 + 2000 \times 0.0009)$$

$$n = 2000 / (1 + 1.8)$$

$$n = 2000 / 2.8$$

$$n \approx 714.29$$

Answer: The sample size should be **715 students**.

Effect of Margin of Error on Sample Size

Margin of Error	Sample Size ($N = 2,000$)
10%	95
5%	334
3%	715
1%	1,667

Key insight: Smaller margin of error requires larger sample size.

Part 9: Parameter vs. Statistic; Probability vs. Nonprobability Sampling

Parameter vs. Statistic

Term	Definition	Symbol	Example
Parameter	A numerical measure that describes a population	μ (mean), σ (standard deviation), P (proportion)	The average height of ALL Grade 11 students in the Philippines
Statistic	A numerical measure that describes a sample	$\bar{x}$ (sample mean), s (sample standard deviation), $\hat{p}$ (sample proportion)	The average height of 200 randomly selected Grade 11 students

Key difference: Parameters are usually unknown (because measuring the entire population is impractical), while statistics are calculated from sample data and used to estimate parameters.

Probability vs. Nonprobability Sampling

Feature	Probability Sampling	Nonprobability Sampling
Definition	Every member of the population has a known, nonzero chance of being selected	Not every member has a chance of being selected; selection is based on convenience or judgment
Bias	Low (random selection reduces bias)	Higher (researcher's choice may introduce bias)
Generalizability	Results can be generalized to the population	Results may not be generalizable
Statistical inference	Possible	Limited or not possible
Examples	Simple random, systematic, stratified, cluster	Convenience, purposive, quota, snowball
When to use	When representativeness is important	When time, cost, or access is limited; exploratory research

Common Nonprobability Sampling Techniques

Technique	Description	Example
Convenience sampling	Selecting participants who are easiest to reach	Surveying classmates

Technique	Description	Example
Purposive sampling	Selecting participants based on specific criteria	Interviewing only teachers with 10+ years of experience
Quota sampling	Selecting a predetermined number from each group	Getting 50 males and 50 females
Snowball sampling	Participants refer other participants	Studying hidden populations (e.g., informal settlers)

Key Terms

Term	Definition
Random variable	A variable whose value depends on the outcome of a random event
Discrete random variable	A variable with countable values
Continuous random variable	A variable that can take any value within a range
Expected value	The average value of a random variable over many trials
Variance	A measure of how spread out values are from the mean
Standard deviation	The square root of the variance
Normal distribution	A bell-shaped, symmetric probability distribution
Standard normal distribution	A normal distribution with $\mu = 0$ and $\sigma = 1$
Z-score	The number of standard deviations a value is from the mean
Percentile	The value below which a given percentage of data falls
Random sampling	Selecting a sample where every member has an equal chance
Parameter	A numerical measure describing a population
Statistic	A numerical measure describing a sample
Slovin's Formula	$n = N / (1 + Ne^2)$ for determining sample size

Review Questions

1. Differentiate between discrete and continuous random variables. Give two examples of each.

2. A discrete random variable X has the following distribution:

X	$P(X)$
0	0.20
1	0.40
2	0.30
3	0.10

Calculate the expected value $E(X)$.

3. Using the same distribution in Question 2, calculate the variance and standard deviation.

4. Describe the characteristics of a normal distribution.

5. A normal distribution has $\mu = 50$ and $\sigma = 5$. Convert $x = 62$ to a z -score.

6. Using the same distribution ($\mu = 50$, $\sigma = 5$), find the raw score for $z = -1.5$.

7. The heights of a population are normally distributed with $\mu = 170$ cm and $\sigma = 8$ cm. Find the probability that a randomly selected person is taller than 180 cm.

8. Using the same distribution ($\mu = 170$, $\sigma = 8$), find the 95th percentile height.

9. A population has 5,000 members. Using Slovin's Formula with a 5% margin of error, find the required sample size.

10. Application question: A company produces light bulbs with an average lifespan of 1,200 hours and a standard deviation of 100 hours. The lifespans are normally distributed.

- (a) What percentage of bulbs last between 1,100 and 1,300 hours?
- (b) What is the probability that a bulb lasts less than 1,000 hours?
- (c) If the company wants to guarantee that only 2% of bulbs fail before a certain number of hours, what should that guaranteed lifespan be?

Answer Key

1. Discrete random variables have countable values (e.g., number of siblings, number of goals scored). Continuous random variables can take any value within a range (e.g., height, temperature).

2. $E(X) = \sum[x \cdot P(x)]$

X	P(X)	X · P(X)
0	0.20	0
1	0.40	0.40
2	0.30	0.60
3	0.10	0.30
Total	1.00	1.30

Answer: $E(X) = 1.30$

3. $\text{Var}(X) = \sum[x^2 \cdot P(x)] - \mu^2$

X	P(X)	X ²	X ² · P(X)
0	0.20	0	0
1	0.40	1	0.40

X	P(X)	X^2	$X^2 \cdot P(X)$
2	0.30	4	1.20
3	0.10	9	0.90
Total	1.00		2.50

$$\text{Var}(X) = 2.50 - (1.30)^2 = 2.50 - 1.69 = \mathbf{0.81}$$

$$\sigma = \sqrt{0.81} = \mathbf{0.90}$$

Answer: Variance = **0.81**; Standard Deviation = **0.90**

4. A normal distribution is bell-shaped, symmetric about the mean, with mean = median = mode. The total area under the curve is 1. It is defined by μ and σ , and follows the Empirical Rule (68-95-99.7).

5. $z = (62 - 50) / 5 = 12 / 5 = \mathbf{2.40}$

6. $x = \mu + z\sigma = 50 + (-1.5)(5) = 50 - 7.5 = \mathbf{42.5}$

7. $z = (180 - 170) / 8 = 10 / 8 = 1.25$
 $P(Z > 1.25) = 1 - 0.8944 = \mathbf{0.1056}$ or **10.56%**

8. $P(Z < z) = 0.95 \rightarrow z \approx 1.645$
 $x = 170 + (1.645)(8) = 170 + 13.16 = \mathbf{183.16 \text{ cm}}$

9. $n = 5000 / (1 + 5000 \times 0.05^2)$
 $n = 5000 / (1 + 12.5)$

$$n = 5000 / 13.5$$

$$n \approx \mathbf{370.37} \rightarrow \mathbf{371}$$

10.

$$\text{(a)} \ z_1 = (1100 - 1200) / 100 = -1.00; \ z_2 = (1300 - 1200) / 100 = 1.00$$

$$P(-1.00 < Z < 1.00) = 0.8413 - 0.1587 = \mathbf{0.6826} \text{ or } \mathbf{68.26\%}$$

$$\text{(b)} \ z = (1000 - 1200) / 100 = -2.00$$

$$P(Z < -2.00) = \mathbf{0.0228} \text{ or } \mathbf{2.28\%}$$

$$\text{(c)} \ P(Z < z) = 0.02 \rightarrow z \approx -2.05$$

$$x = 1200 + (-2.05)(100) = 1200 - 205 = \mathbf{995 \text{ hours}}$$